

choptuik_ac_bc · theorems12 · github.com/wild8highlander/choptuik_ac_bc

The Center Hierarchy

Theorem: the O1–O5 ladder derives every central amplitude

Residual orders fixed per equation · χ -corrections vanish at $\chi = 0$ · the flat chain is the $\chi = 0$ slice

Author:	Isahk Khamzatovich Isaev
Series:	Monograph No. 18 of the theorems12 cycle (supplementary series)
Machine:	sympy_center.py · SymPy, exact ($\kappa = 2$)
Verdict:	einstein_direct/results/ center_hierarchy.json
Status:	PROVEN EXACTLY (hierarchy + χ - corrections)

Abstract

This supplementary monograph records the center hierarchy theorem: the central (near-center) expansion of the corrected Einstein–scalar system is organized in a rigid ladder O1–O5 in which each order fixes the next amplitude. The ansatz $r = R1\xi + R3\xi^3$, $\Phi = P0 + P2\xi^2 + P4\xi^4$, $\omega = W0 + W2\xi^2$, $m = M3\xi^3$ reduces the twelve equations to residual orders that are nonzero exactly at fixed powers of ξ — SC at $\{0, 2, 4\}$, UV at $\{1, 3\}$, TH at $\{0, 2, 4\}$, Mdef at $\{1, 3\}$, C1/C2 at $\{0, 1, 2, 3, 4\}$ — and the order-by-order solution yields the exact hierarchy O1–O5.

The χ -corrections (for the shifted center $x = \xi + x_s(y)$) are computed exactly and vanish identically at $\chi = dx_s/dy = 0$ — the flat chain of theorems No. 1–6 is the $\chi = 0$ slice of a consistently derived family, not an isolated guess. The hierarchy table, the χ -correction forms, corollaries, honest caveats and the reproduction protocol are included.

Keywords: center expansion, hierarchy O1–O5, χ -corrections, near-center ansatz, Einstein–scalar system, SymPy, machine theorem.

Table of Contents

1. Introduction.....	1
2. Notation and Setting.....	1
3. Statement of the Theorem.....	2
4. Sketch of the Machine Proof.....	2
5. Numerical Results.....	3
6. Corollaries.....	3
7. Honest Caveats.....	4
8. Reproducibility.....	4
References.....	5

Note: the Table of Contents is generated via field codes. After editing, refresh the page numbers: right-click the TOC and select "Update Field".

1. Introduction

This supplementary monograph records the center hierarchy theorem — the machine result of the center campaign (machine `sympy_center.py`, verdict `center_hierarchy.json`, LEDGER §10). The central ansatz is the near-center expansion of the double-null fields around the worldline of the center; the question was whether the expansion is a consistent hierarchy (each order determined) or an ansatz with free functions left undetermined.

The machine answer: the hierarchy is rigid. The residual orders are fixed per equation, and the O1–O5 ladder determines t_0' , R_1' , d_0' , M_3 and P_2'' successively from the lower amplitudes. Moreover, the χ -corrections for a shifted center are exact and vanish at $\chi = 0$ — the flat chain used throughout the symbolic block is the $\chi = 0$ slice of the same family. This theorem is the derivation backbone that the clock-pair, factorization and dd-closure theorems silently reuse.

2. Notation and Setting

The center ansatz in the comoving coordinate $\xi = x - x_s(y)$: $r = R_1(y)\xi + R_3(y)\xi^3$, $\Phi = P_0(y) + P_2(y)\xi^2 + P_4(y)\xi^4$, $\omega = W_0(y) + W_2(y)\xi^2$, $m = M_3(y)\xi^3$, with $\chi = dx_s/dy$ the center-shift slope. Substituting into the twelve equations of the corrected system and expanding in ξ gives the residual orders of Table 18.1; the hierarchy is the order-by-order system O1–O5.

Table 18.1. The nonzero residual orders per equation ($\chi = 0$ layer)

Equation	Nonzero ξ -orders	Reading
SC	0, 2, 4	even sector of the scalar equation
UV	1, 3	odd sector of the u-equation
C2	0, 1, 2, 3, 4	the full v-constraint ladder
TH	0, 2, 4	even sector of the trace equation
C1	0, 1, 2, 3, 4	the full u-constraint ladder
Mdef	1, 3	odd sector of the mass defect

3. Statement of the Theorem

Theorem 18 (the center hierarchy). For the corrected Einstein–scalar system at $\kappa = 2$:

- (i) the residual orders are exactly those of Table 18.1 — no equation carries a residual outside its sector, and the parity structure is machine-exact;
- (ii) the order-by-order system is the rigid ladder O1–O5: O1 fixes $t_0' = 3P_2 - P_0'R_1'/R_1$; O2 fixes $R_1' = R_1 \cdot W_0'$; O3 fixes $d_0' = M_3 \cdot e^{\{2W_0\}}/R_1^3 + W_2 - P_0'^2/2$; O4 fixes the M_3 series; O5 fixes P_2'' ; each right-hand side involves only lower or already-determined amplitudes;
- (iii) the gauge condition of the mass defect is $M_{\text{def}}_{\xi 1}: R_1^3 e^{\{-2W_0\}}/2 - R_1/2 = 0$, and the linear links close: $UV_{\xi 1}$ relates R_1'' to $M_3 \cdot e^{\{2W_0\}}/(2R_1^2) - 3R_3/2$, and $C1_{\xi 0} = R_1 W_0'/2 - R_1'/2$ ties the logarithmic potential to the ruler;
- (iv) the χ -corrections are exact: $SC_{\xi 0}$, $SC_{\xi 1}$, $C2_{\xi 0}$, $C2_{\xi 1}$, $TH_{\xi 0}$ are explicit polynomials in $\chi = dxs/dy$ (quoted by the machine), and every one of them vanishes identically at $\chi = 0$ — the flat chain is the $\chi = 0$ slice of the derived family;
- (v) the numeric subcritical check reproduces the flat-chain booking at machine zero: the hierarchy solution at the booked amplitudes carries residuals at the 10^{-14} level of the float64 verification layer.

4. Sketch of the Machine Proof

The machine substitutes the ansatz into the twelve equations, expands in ξ to the needed order, and collects the coefficient polynomials per equation; the parity check is exact (zero coefficients outside the sector). The O1–O5 ladder is then read off as the triangular system; each right-hand side is simplified symbolically. The χ -layer repeats the substitution with $\xi = x - xs(y)$ and expands in χ ; the correction forms are explicit and linear in χ at the first orders. The verification layer numerically evaluates the hierarchy at the booked chain amplitudes and confirms the residual floor.

Table 18.2. The hierarchy O1–O5 (right-hand sides, $\kappa = 2$)

Order	Solves for	Right-hand side
O1	$t0'$	$3P2 - P0' \cdot R1' / R1$
O2	$R1'$	$R1 \cdot W0'$
O3	$d0'$	$M3 \cdot e^{\{2W0\}} / R1^3 + W2 - P0'^2 / 2$
O4	$M3$	$(2R1^3W2 - 7R1^2R3 + R1R1'^2 + R3e^{\{2W0\}}) \cdot e^{\{-2W0\}} / 2$
O5	$P2''$	$2(-(-4P2R3 + P0'R3' + P2'R1')R1 + 10P4R1^2 + R3P0'R1') / R1^2$

5. Numerical Results

 Table 18.3. The χ -corrections (leading forms, machine-exact)

Correction	Form (leading in χ)
SC_ ξ_0	$((6P2\chi^2 - 6P2 + P0'')R1 + 2P0'R1') / (4R1)$
SC_ ξ_1	$-(P2\chi'' + 3P2'\chi)R1/2 - P2R1'\chi - R3P0'\chi / R1$
C2_ ξ_0	$R1W0'\chi/2 - R1W0'/2 - R1\chi''/4 - R1'\chi/2 + R1'/2$
TH_ ξ_0	$-M3e^{\{2W0\}} / (2R1^3) + W2\chi^2/2 - W2/2 + \dots$

Every correction is a polynomial in χ and its derivatives; at $\chi = 0$ each reduces to the flat-chain residual, which the constraint system then annihilates. The numeric subcritical layer confirms the same at the float64 floor: the hierarchy evaluated at the booked amplitudes carries residuals $\leq 10^{-13}$ across all twelve equations.

6. Corollaries

Corollary 18.1 (the flat chain is derived, not guessed). Theorems No. 1–6 operate on the flat chain; this theorem shows the chain is the $\chi = 0$ slice of a consistent family — the

ansatz behind the whole symbolic block is machine-derived, and the χ -direction is available for future extensions.

Corollary 18.2 (the ladder is rigid). Any future central ansatz must pass the same parity test (Table 18.1) and reproduce the O1–O5 ladder; a scheme that leaves a free function at a fixed order is outside the class.

Corollary 18.3 (the books are hierarchy-level facts). The ring $4/3$ and the clock $P2h/T0h = 1/3$ of the P4 theorem (No. 19) are consequences of this hierarchy's core — the machine derives them, it does not postulate them.

7. Honest Caveats

The hierarchy is exact at $\kappa = 2$ in the T1c source form; the κ -dependence of the ladder is handled by the holonomy theorem (No. 7) at the book level but not re-derived here order by order. The χ -layer is computed to the first nontrivial orders; higher χ -orders are not exhausted. The ansatz is central (near-center) — it says nothing about the far zone, and the matching to the PDE machine is outside the theorem. Finally, the numeric subcritical check is a float64 verification; its floor 10^{-13} – 10^{-14} is a representation limit, not a symbolic statement.

8. Reproducibility

Table 18.4. Reproduction protocol for Theorem 18

Step	Command	Result
1	<code>python3 sympy_center.py</code>	results/center_hierarchy.json (hierarchy + χ -layer)
2	<code>python3 -m pytest tests/ -q</code>	58 passed (center tests included)

The machine runs in seconds (deterministic, offline). The data correspond to commit 5b256cd; an autonomous copy lives in `einstein_direct/open9/machines/`.

References

- [1] Repository `wild8highlander/choptuik_ac_bc`: `einstein_direct/sympy_center.py`; `einstein_direct/results/center_hierarchy.json` (LEDGER §10).
- [2] Monographs No. 1, No. 3, No. 5, No. 19 of the theorems12 cycle.